

```
int p = 14;
int q = 6;
int r = -9;
double d = 3.75;
boolean flag1 = false;
boolean flag2 = true;
String name1 = "Store";
String name2 = "Shore";
```

1A)

1. $p \% q * 2 + r$

$14/6*2+(-9) \rightarrow 14/6=2 \rightarrow 2*2+(-9) = -5$ (int)

2. $(p + r) / q$

$(14+(-9))/6 \rightarrow \% = 0$ (int)

3. p / q

$14/6 \rightarrow 2$ (int)

4. $d * 4 - p \% q$

$3.75*4-14/6 \rightarrow 3.75*4=15.0, 14/6=2 \rightarrow 15.0-2 = 13.0$ (double)

5. $-r + p / q * q$

$-(-9)+14/6*6-(-9) + 14/6*6 \rightarrow 9+4/6*6 \rightarrow 14/6=2, 2*6=12 \rightarrow 9+12= 21$ (int)

1B)

6. $p < q \parallel r < 0$

$14 < 6 \parallel -9 < 0 \rightarrow \text{false} \parallel \text{true} = \text{true}$ (boolean)

7. $p / q == 2$

$14/6=2 \rightarrow 2==2 = \text{true}$ (boolean)

8. $d > 3.75$

$3.75 > 3.75 = \text{false}$ (boolean)

9. $r != p - 23$

$14-23=-9 \rightarrow -9! = -9 = \text{false}$ (boolean)

10. $\text{name1.compareTo(name2)} < 0$

S = S

t > h

"Store".compareTo("Shore") > 0, "Store".compareTo("Shore") < 0 = false(boolean)

1C)

11. flag1 && flag2

false && true = false(boolean)

12. flag1 || flag2

false||true = false(boolean)

13. !flag2

flag2=true -> !true = false(boolean)

14. flag2 && !flag1

true && !false -> true && true = true(boolean)

15. (p > 10) || (q > 10)

(14>10) || (6>10) -> true||false = true(boolean)

16. (p > 10) && (q > 10)

(14>10) &&(6>10) -> true && false = false(boolean)

17. !(p > 10 && q > 10) — evaluate directly, then rewrite using De Morgan's Law and confirm the two match

(14>10)&&(6>10) -> true && false -> false -> !(false) = true(boolean) -> apply Law ->

!(14>10) || !(6>10) -> !true || !false -> false||true -> true(boolean)

18. p < q || r < q && flag2 — careful with precedence (&& binds tighter than ||)

14 < 6 || (-9 < 6 && true), false || (true && true) -> false||true = true(boolean)

19. !(flag1 || flag2) == (!flag1 && !flag2)

!(false || true) == (!false && !true) -> false || true=true !true=false, !false && !true true && false = false, false == false = true(boolean)

20. Short-circuit trace: given int div = 0;, does (div != 0) && (20 / div > 1)

throw a division-by-zero error? Explain why or why not in one sentence.

(0 != 0) && (20 / 0 > 1) -> 0!=0 -> false -> No it does not throw a division-by-zero error.

explanation: Once the left side is false java does not evaluate the right side. So it never does 20/0

1D)

21. 3 * 8 <= 4 + 2 * (9 - 4)

$3 * 8 \leq 4 + 2 * 5 \rightarrow 24 \leq 4 + 10 \rightarrow 24 \leq 14 \rightarrow ((3 * 8) \leq (4 + (2 * (9 - 4)))) = \text{false}(\text{boolean})$

22. $p + r > q * 2 \ \&\& \ d < 4.0$

$14 + (-9) > 6 * 2 \ \&\& \ 3.75 < 4.0 \rightarrow 5 > 12 \ \&\& \ 3.75 < 4.0 \rightarrow \text{false} \ \&\& \ \text{true} \rightarrow (((p + r) > (q * 2)) \ \&\& \ (d < 4.0)) = \text{false}(\text{boolean})$

23. $!flag1 \ || \ flag2 \ \&\& \ flag1$

$!false \ || \ true \ \&\& \ false \rightarrow true \ || \ true \ \&\& \ false \rightarrow true \ || \ false \rightarrow ((!flag1) \ || \ (flag2 \ \&\& \ flag1)) = \text{true}(\text{boolean})$

24. $15 - 2 * 4 > 5 \ || \ false$

$15 - 8 > 5 \ || \ false \rightarrow 7 > 5 \ || \ false \rightarrow true \ || \ false \rightarrow (((15 - (2 * 4)) > 5) \ || \ false) = \text{true}(\text{boolean})$

25. $p \% 2 == 0 \ || \ q \% 2 == 0 \ \&\& \ r \% 2 ==$

$14 \% 2 == 0 \ || \ 6 \% 2 == 0 \ \&\& \ -9 \% 2 == 0 \rightarrow 14 \% 2 = 0, 6 \% 2 = 0, -9 \% 2 = -1 \rightarrow 0 == 0 \ || \ 0 == 0 \ \&\& \ -1 == 0 \rightarrow true \ || \ true \ \&\& \ false \rightarrow true \ || \ false \rightarrow ((p \% 2 == 0) \ || \ ((q \% 2 == 0) \ \&\& \ (r \% 2 == 0))) = \text{true}(\text{boolean})$